An energy budget

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Connect work, motion, and heating. State a conclusion that fits the calculation's limits.

SUCCESS CRITERIA - I can...

1. Keep all terms in consistent energy units.
2. Distinguish mechanical-energy change from total-energy loss.
3. State what you treat as true when you account for the energy.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A **0.3 kg** object moving at **5 m/s** stops through friction over **0.7 m**. **Find energy transformed** and average opposing-force magnitude.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $E_i + E_{in} = E_f + E_{out}$.
i/f = initial/final stored energy; in/out = transfers across the chosen system boundary. Account for every form.

Worked reasoning: An input of **100 J** gives **60 J** useful motion, **25 J** thermal transfer and **15 J** sound/other transfers. The account balances because **60**+**25**+**15**=**100 J**.

Your turn: Draw proportional energy arrows and **calculate the useful fraction**.

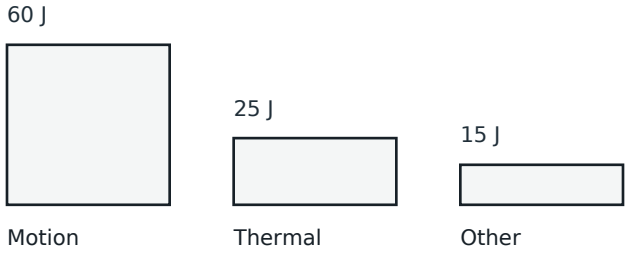
Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

A modeled system starts with **100 J**. It ends with **60 J** motion energy, **25 J** thermal energy and **15 J** sound/deformation. **Draw an energy account** and **check the total**.



Bar height represents energy. Read the value and unit above each bar.

Marzano check - circle: ① ② ③

04 Apply the physics

A simulated **1000 kg** probe has K = **8 × 10⁶ J** and gravitational U = **-6 × 10⁶ J**, with U = **0** at infinity. **Find total mechanical energy**. Drag then removes **3 × 10⁶ J**. **Complete the energy account** and discuss whether the final state is bound in a calculation with only the central body's gravity acting.

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

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Continue down each column.

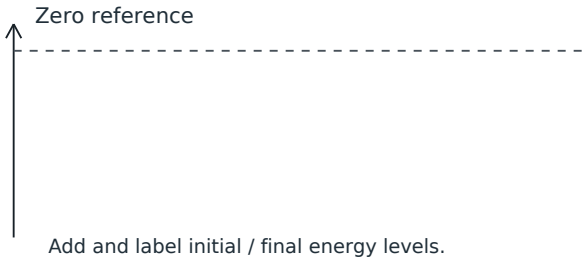
05 Explain your reasoning

What information is missing if an impact report gives only total kinetic energy and says all of it heats the ground?

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

With GMm=**192 J·m**, move an object from r=**4** to r=**8 m**.
Find work by gravity using U=-GMm/r.
Label the distances or energy levels used in your reasoning.



Add and label initial / final energy levels.

Marzano check - circle: 1 2 3

07 Exit check

A **2 kg** cart descends **3 m** from rest with g=**10**. Losses are **12 J**.
1. **Find bottom speed**.
2. **Use the energy** diagram to **show the quantities you compare**; **label units** and any unknowns.

Initial **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

A device takes **600 W** for **5 s** and raises a **10 kg** load **12 m** using g=**10**.
1. **Find useful work**, efficiency and remaining energy.
2. **Use the energy** diagram to **show the quantities you compare**; **label units** and any unknowns.

Initial **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3